

AS IS, WHERE IS

Does a tabular foundation model need
the table cleaned first?

TabPFN-3.5, zero-shot, on the raw Kaggle *Blue Book for Bulldozers* auction table

**NO WARRANTY.
NO CLEANUP.**

Kaggle table, as
published

THE BET

Price the auction table as is, where is

THE CLAUSE

Used heavy equipment is auctioned “as is, where is”: no warranty, no cleanup.

THE QUESTION

Can a tabular foundation model price under the same clause, taking the data as it comes?

THE CLAIM, PRE-REGISTERED

Given the 2013 Kaggle table exactly as published, zero-shot TabPFN-3.5 prices the next four months of sales as well as a LightGBM given 2013-style cleaning and feature engineering.

THE DATA • 401,125 TRAINING SALES, 1989–2011

Kaggle's table is dirty as published

1000

recorded as YearMade for 9.5%
of sales

64%

of hour meters missing; another
18% read zero

35 of 51

predictor columns more than
half empty

5,218

model IDs, plus free-text model
descriptions

3/14/2012 0:00

the sale date, as a text string

0

cleaning steps for TabPFN-3.5

LightGBM gets the cleaning; TabPFN-3.5 does not

BASELINE · LIGHTGBM

2013-style cleaning and features

- YearMade 1000, zero hours → missing
- Date → year, month, weekday, day count
- Machine age at sale; 55 features
- Early stopping on the latest 10%

TABPFN-3.5 · ZERO-SHOT VIA THE API

The table as published

- 51 predictor columns, as in the CSV
- Date as the text “3/14/2012 0:00”
- Default settings, no tuning
- All 401,125 sales as context

Context · 401,125 sales, Jan 1989 – Dec 2011

Scored · 11,573 sales, Jan–Apr 2012, against Kaggle’s published prices

Raw and zero-shot, TabPFN-3.5 beats the baseline

0.2200

TabPFN-3.5 · raw table · zero-shot

0.2325

LightGBM · 2013-style cleaning and features

RMSLE on the 11,573 sales of Jan–Apr 2012, lower is better. Paired difference **−0.01245**, 95% CI −0.01480 to −0.01021: the whole interval below zero.

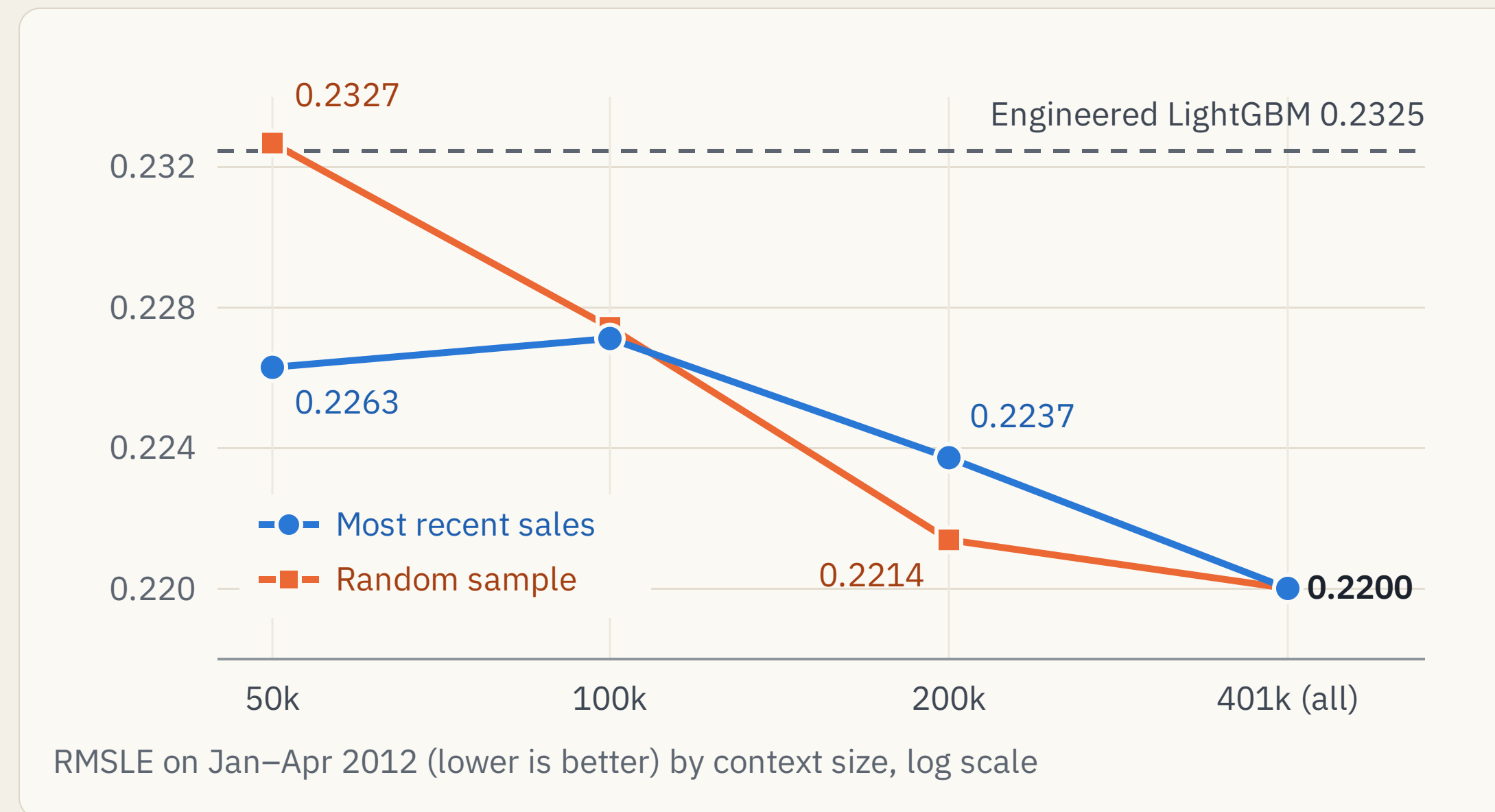
10 of the 11 distinct TabPFN-3.5 configurations scored below the LightGBM.

Four hypotheses survive, one is refuted

#	Hypothesis	Result	Verdict
H1	Raw TabPFN-3.5 is no worse than the LightGBM	−0.01245, CI below zero	Corroborated
H2	Cleaning or the appendix gains under 0.005	+0.00175 and +0.00267	Corroborated
H3	Recent sales beat a random sample, equal size	yes 50k · tie 100k · no 200k	Refuted
H4	The 80% interval covers 75–85%, 4 months out	82.0% coverage	Corroborated
H5	RMSLE below the 2013 top-10% mark, 0.2534	0.2200, mirror split, no rank	Corroborated

RMSLE on Jan–Apr 2012; differences are paired bootstraps (10,000 resamples, 95% CI); H1 and H2 use a 0.005 margin. A script writes the verdicts; an independent AI agent recomputed every number. H5 compares across test sets, so it is indicative, not a leaderboard rank.

Recency helps at 50k sales and hurts at 200k



50k · recent wins

−0.00639 (CI −0.00839 to −0.00440)

100k · a tie

−0.00032 (CI −0.00215 to +0.00152)

200k · random wins

+0.00234 (CI +0.00079 to +0.00389)

Like choosing comparables: with few, recency carries the market; with many, coverage of models matters more. An appraiser's reading, not tested.

Cleaning helps, but by less than the margin

Raw table (primary run)

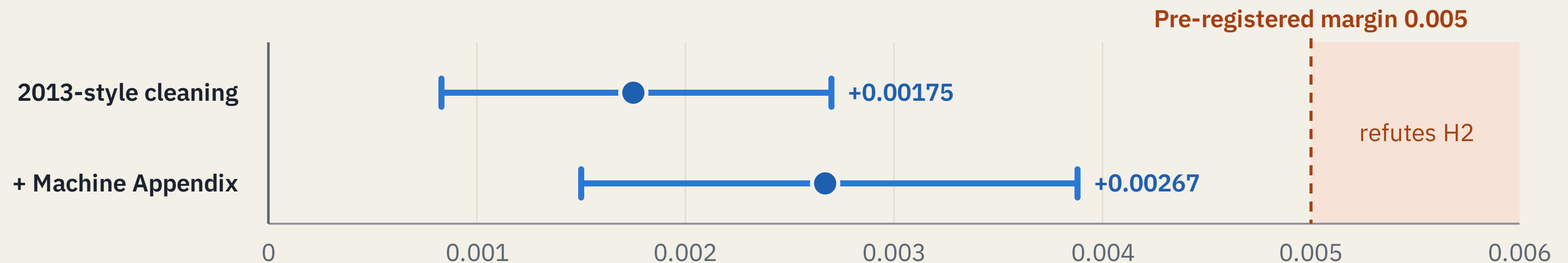
0.2200

2013-style cleaning

0.2183

Raw + “corrected” Machine Appendix

0.2173

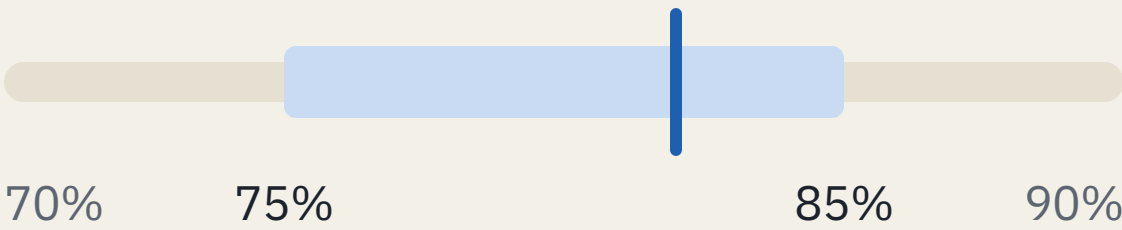


RMSLE improvement over the raw table, with 95% paired-bootstrap intervals. Both gains are real (intervals above zero) and below the margin. The 2013 report that the appendix did not help is not reproduced in its strong form.

The 80% interval holds four months ahead

82.0%

of realized prices fell inside the predicted 80% interval [q10, q90]



Shaded: the pre-registered band, 75–85%

WHERE THE REALIZED PRICES FELL

Predicted quantile	q10	q25	q50	q75	q90
Nominal share	10%	25%	50%	75%	90%
Realized at or below	7%	19%	44%	71%	89%

All five quantiles sit a little low: realized prices ran **2.6% above** the forecast, in geometric mean.

The cause was not tested. A general price rise is not supported (the Jan–Apr median was US\$27,000 in both 2011 and 2012); seasonality is compatible.

EXTERNAL REFERENCE · SAME BASIS

Ahead of a published model on the same split

Model · trained on sales through 2011, scored on Jan–Apr 2012	RMSLE
TabPFN-3.5, raw table + Machine Appendix	0.2173
TabPFN-3.5, raw table (primary run)	0.2200
LightGBM, 2013-style cleaning and features (this study)	0.2325
Random forest, tuned, Parr & Howard (<i>The Mechanics of Machine Learning</i>)	0.2396

A point comparison only: their predictions are not public, so no paired test. The 2013 winner (0.2291) was scored on the hidden May–Nov 2012 test set, which Kaggle no longer scores, and the public leaderboard on this split is contaminated: 108 of 475 teams show 0.0.

Quoted, measured and reproducible

732,938 tokens

quoted by the server and measured for the cleaning and appendix runs, exactly (context block: 326,828 both)

Identical

predictions from one configuration run on two days:
0 USD gap in the mean and five quantiles

Half the time

fast mode on all sales: RMSLE 0.2205 vs 0.2200, no detectable loss (813 s vs 1,654 s)

2.36M tokens

the blocks of 1 Oct 2026 (all but pilot and primary), under the 5M daily limit; thinking mode was worse at 50k

LIMITS AND LINKS

One period, one market, one plain baseline

- One evaluation period (Jan–Apr 2012), one table, one market
- A plain LightGBM: fixed settings, no per-group models, no blending
- Kaggle no longer scores late entries; the test-set file is in the repo
- H4 bias of 2.6%: cause not established

REPOSITORY · APACHE-2.0

github.com/abenvenho/tabpfn-bluebook-bulldozers

Pre-registration, code, results, figures and the 9-page paper

ENTRY 1

github.com/abenvenho/tabpfn-itbi-sp

São Paulo real-estate transfer prices